Examiner only

4. *(a)* Green plants capture a small percentage of solar energy during the process of photosynthesis. The table below shows how the absorption of light by a green plant depends on the wavelength of the visible light.

Wavelength of visible light (nm)	Absorption (%)
400	85
450	100
500	2
550	12
600	18
650	35
700	75

(i) Describe how the absorption of light depends on the wavelength of visible light. [3]

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(ii) Electromagnetic waves travel at 3×10^8 m/s in space.

Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the frequency of light which is most strongly absorbed. [3]

($1\,\text{nm} = 1 \times 10^{-9}\,\text{m}$)

Frequency = ... Hz

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(b) The table shows what happens to the energy taken in each day by organisms in the food chain below.

grass ⟶ grasshopper ⟶ shrew ⟶ fox

Organisms	Energy per day (MJ)		
	As waste	Released during respiration	Used for growth
grass	6	10	8
grasshopper	14	22	3
shrew		26	4
fox	24	32	4

(i) The shrew releases 60% of its energy during respiration and for growth. Calculate how much energy is released by the shrew as waste. [2]

Energy as waste = ... MJ

(ii) Explain why the amount of energy released during respiration by each organism increases through the food chain. [2]

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(iii) Sometimes plant seeds will stick to the coat of a fox. Explain how this benefits the plant. [2]

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